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Technical Specification

O-RAN White Box Hardware Working Group
Hardware Reference Design Specification for Fronthaul Gateway

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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

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where:

x the first digit is incremented for all changes of substance, i.e., technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).

y the second digit is incremented when editorial only changes have been incorporated in the document.

z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies a hardware reference design for the fronthaul gateway. The fronthaul gateway can be used to terminate O-RAN fronthaul user, control, synchronization, and management planes for option 7-2x and to translate the signals to and from a number of CPRI interfaces (option 8). It can further multiplex or demultiplex option 7-2x fronthaul signals to or from a number of O-RUs. This version of the specification only supports option 7-2x to option 8 conversion and vice versa in addition to the transport of other traffic streams.

In the main body of this specification (in any "chapter") the information contained therein is informative, unless explicitly described as normative. Information contained in an "Annex" to this specification is always informative unless otherwise marked as normative.

1.2 References

[1] ORAN-WG7.DSC.0-V02.00 Technical Specification, 'Deployment Scenarios and Base Station Classes for White Box Hardware'.

[2] ORAN-WG4.CUS.0-v04.00 Technical Specification, 'O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification'.

[3] O-RAN Architecture Description, O-RAN-WG1-O-RAN Architecture Description - v01.00.00

[4] CPRI interface specification V 7.0. http://cpri.info/spec.html

[5] O-RAN Hardware Reference Design Specification for Indoor Picocell FR1 with Split Architecture Option 8 v1.0

1.3 Definitions and Abbreviations

1.3.1 Definitions

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x
5G	Fifth-Generation Mobile Communications
ASIC	Application Specific Integrated Circuit
BS	Base Station
CU	Centralized Unit as defined by 3GPP
DL	Downlink
DPLL	Digital Phase Lock Loop
DSP	Digital Signal Processor
DU	Distributed Unit as defined by 3GPP
FFT	Fast Fourier Transform
FH	Fronthaul
FHGW	Fronthaul Gateway
$\text{FHGW}_{x \to y}$	Fronthaul Gateway that can translate FH protocol from an O-DU with split option x to an O-RU with split option y, with currently available option 7-2→8.
FPGA	Field Programmable Gate Array
GbE	Gigabit Ethernet
GNSS	Global Navigation Satellite System
IEEE	Institute of Electrical and Electronics Engineers
I/O	Input/Output
LTE	Long Term Evolution
MAC	Media Access Control
O-CU	O-RAN Centralized Unit as defined by O-RAN
OCXO	Oven Controlled Crystal Oscillator
O-DUx	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2 (as defined by WG4) or 8
O-RUx	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2 (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DU_x
PCIe	Peripheral Component Interface Express
POE	Power over Ethernet
RAN	Radio Access Network
RF	Radio Frequency
RU	Radio Unit as defined by 3GPP
RX	Receiver

SFP	Small Form-factor Pluggable
SFP+	Small Form-factor Pluggable Transceiver
SoC	System On Chip
TX	Transmitter
UL	Uplink
USB	Universal Serial Bus
WG	Working Group

Chapter 2 Deployment Scenarios

O-RAN Working Group 7's deployment scenario specification [1] shows the split architecture as below. The Fronthaul Gateway (FHGW) in this case aggregates multiple O-RUs to a single location consisting of O-DUs. The fronthaul gateway provides protocol translation, packet switching and aggregation capabilities.

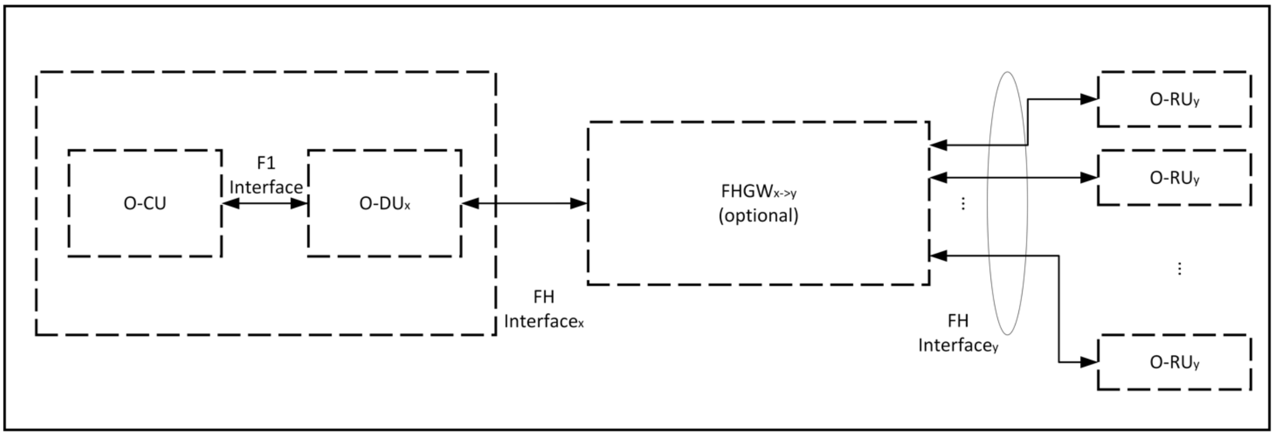

Figure 1: Base Station with Split Architecture

This document describes scenarios where the FHGW provides protocol translation in addition to the switching and aggregation capabilities.

2.1.1 Deployment Scenario -1

The FHGW (Fronthaul Gateway) may be placed between the O-DU and RU (Radio Unit) with the following O-RAN specified interfaces:

- The interface between O-DU and FHGW is Open Fronthaul (Option 7-2x).
- The interface between FHGW and RU is an LLS option specified by O-RAN.
- The interface between FHGW and RU may not support Open Fronthaul (Option 7-2x).

Figure 2 below depicts the deployment of the FHGW using O-RAN specified interfaces.

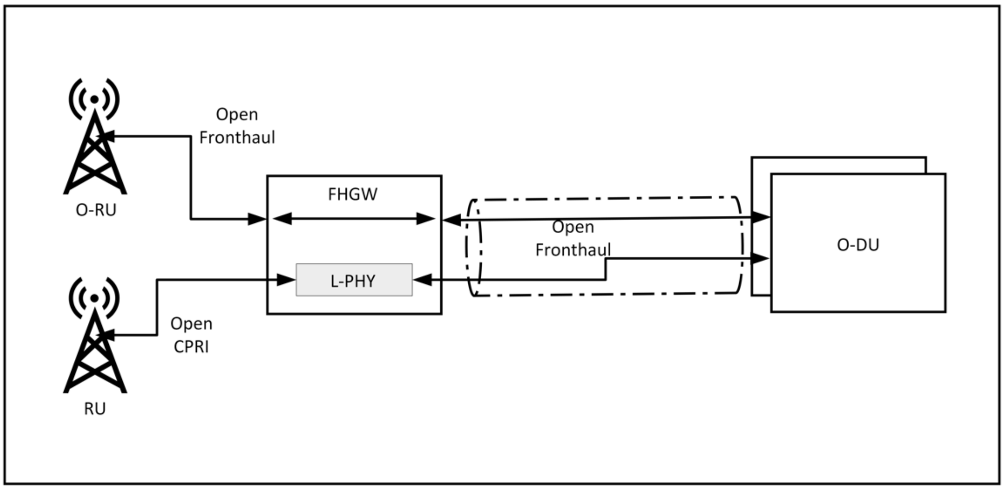

Figure 2: Deployment of O-DU and RU using FHGW

The Low PHY function in the fronthaul gateway converts the CPRI/low level split interface between the RU and Open Fronthaul (Option 7-2x). The fronthaul gateway is deployed near the RU physically and gets connected to the DU using a packet-based network.

The FHGW gateway has radio processing capabilities to do the protocol translation between CPRI/low level split interface and Open Fronthaul (Option 7-2x) and does the translation for the management plane also. The non-O-RAN-RU along with FHGW appears as a single/multiple O-RU(s) to the O-DU.

The RUs and O-RUs are directly connected to the FHGW. The O-DUs are connected to FHGW over a network. The network between FHGW and O-DU could be a point-to-point, mesh or a ring topology. In some cases, the FHGWs provide transport to non-fronthaul traffic also, namely, midhaul, backhaul or other traffic. So FHGW may have non-fronthaul ethernet interfaces also.

2.1.1.1 Point to point network

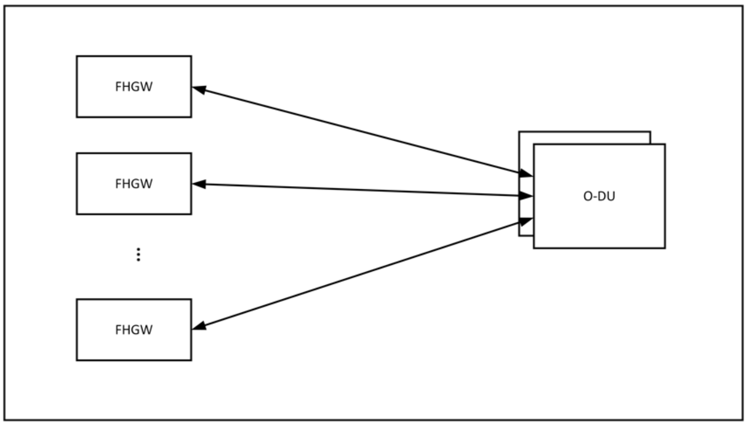

Figure 3: Point to point topology for Fronthaul Gateway

Each fronthaul gateway gets connected to the O-DUs directly. These connections might be direct physical connections or through a packet network. When FHGWs connect through other network elements to the O-DUs, they form a mesh network.

2.1.1.2 Ring network

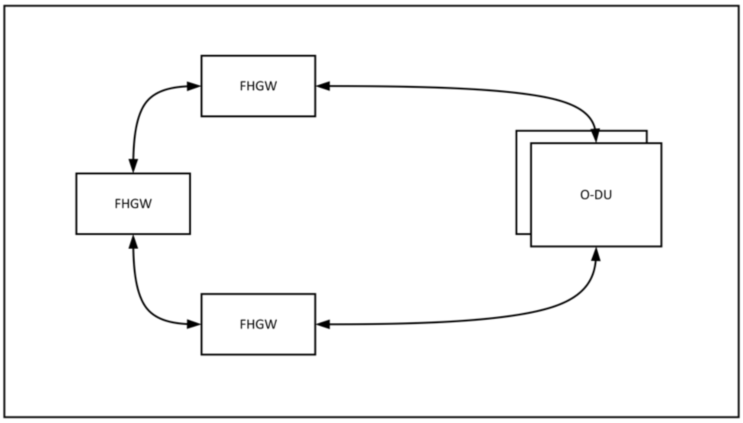

Figure 4: Ring topology for Fronthaul Gateway

FHGWs form a ring topology and get connected to the O-DUs. In this case, the Fronthaul Gateways need two interfaces for connectivity towards the O-DUs

2.1.2 Low PHY

The Low PHY module in fronthaul gateway implements the below functionalities.

1) FFT/IFFT (Lower-PHY DL/UL)
2) PRACH detection (Lower-PHY UL)
3) Handling of C-Plane/M-plane messages
4) Timing and synchronization of RU
5) eCPRI framing/de-framing and switching
6) CPRI framing/de-framing and switching
7) CPRI to eCPRI conversion
8) I/Q compression on eCPRI and CPRI links

Doing LowPHY functionality on the FHGWs and converting the low-level split interface to O-RAN fronthaul interface enables connecting legacy RUs to O-DUs. This enables brownfield deployments to leverage the benefits of O-RAN based disaggregated and open networks.

2.1.3 Timing

The packet based fronthaul network is compliant with the timing requirements described by WG4 CUS specification [2]. The packet-based network uses SyncE and PTP to provide frequency, phase and Time-of-Day to necessary endpoints. It can derive these parameters from satellite-based sources also. The legacy RUs derive frequency and phase through the CPRI/low level split interface connected to the fronthaul gateway. CPRI/low level split interface transports frequency and phase through mechanisms difference from SyncE and PTP. The scheme is specified in the CPRI v7.0 [4] specification and is not vendor specific.

2.1.4 Attributes of FHGW$_{7\text{-}2\rightarrow 8}$

Table 1: Attributes of FHGW$_{7\text{-}2\rightarrow 8}$

Attributes	Values or Assumptions
Functionalities	Packet switching, Radio interface translation
Timing Capabilities	Compliant to S-Plane protocol specification in O-RAN-WG4-CUS.0-v04.00 specification
Deployment	Outdoor, near Outdoor Macrocell
Interfaces	Low level split Interfaces to connect to legacy RU Ethernet interfaces to connect to O-DU Ethernet interfaces to connect to O-RU Timing interfaces

Chapter 3 Hardware Architecture and Requirements

Fronthaul gateways are deployed near the Radio units where it aggregates multiple Radio Units. In case of brownfield RUs, fronthaul gateway converts the split 8 signals to split-7-2x specified by O-RAN. In macrocell deployment scenarios, the FHGW will cater to xHaul needs as well. So, it will handle diverse traffic streams.

In this case the fronthaul gateway has digital signal processing capabilities as well as packet switching capabilities. In case of a FHGW, the combination of RU and FHGW acts as an O-RU.

3.1 FHGW$_{7\text{-}2\rightarrow 8}$ Hardware Architecture

This section describes the hardware architecture of a fronthaul gateway which supports legacy interfaces towards RUs and provides protocol translation to enable them to connect to O-DUs. In addition to that Fronthaul Gateway provides transportation capabilities for O-RAN fronthaul traffic as well as other types of traffic.

3.1.1 FHGW$_{7\text{-}2\rightarrow 8}$ Hardware Architecture Diagram

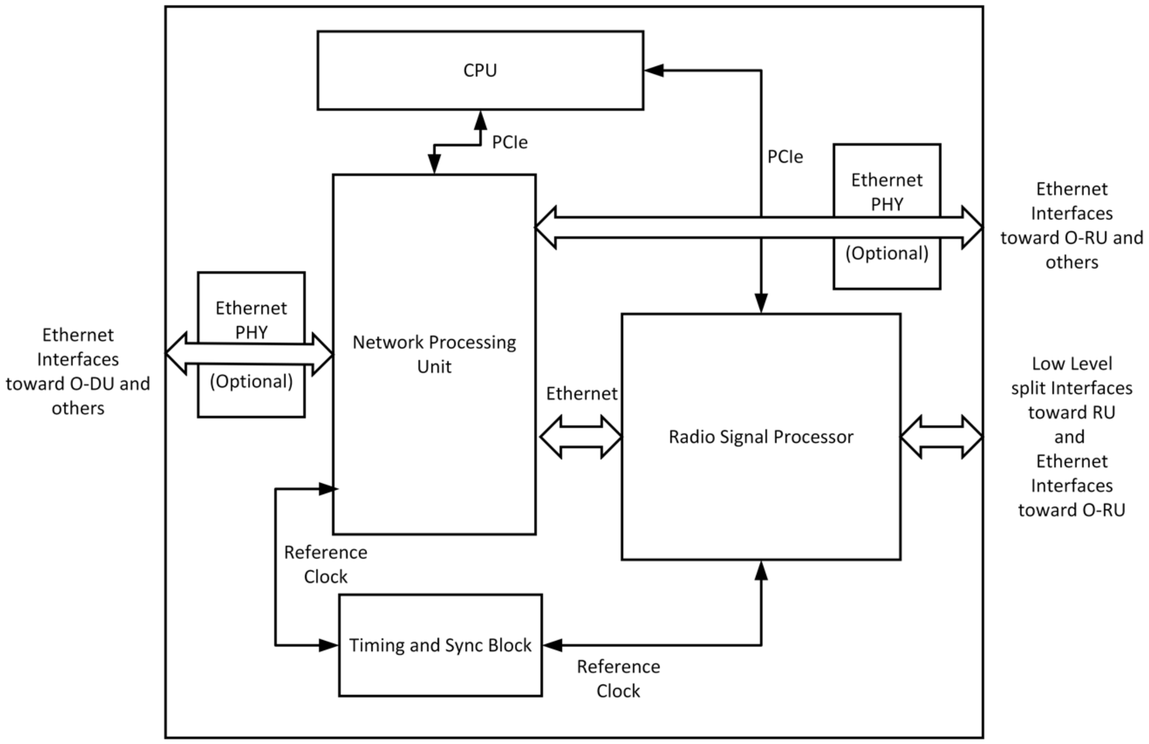

Figure 5: Fronthaul Gateway Hardware Architecture Diagram

3.1.2 FHGW$_{7\text{-}2\rightarrow 8}$ Functional Module Description

This section describes the functional components of a fronthaul gateway which does protocol translation from non-O-RAN fronthaul interface towards RUs to O-RAN fronthaul interface (split 7-2) towards O-DUs. This fronthaul gateway supports O-RAN fronthaul interface (split 7-2) towards O-RUs and provides transportation for that traffic towards O-DUs. Fronthaul gateway can handle non-Fronthaul traffic also. The selection of specific hardware components for these modules is a product design task. The descriptions and requirements are given below.

Network Processing Unit

The network processing unit provides the packet transport functions of the FHGW.

Radio Signal Processor/Accelerator

Radio Signal processor/Accelerator does the protocol translation capabilities required to translate from the low-level split-8 to O-RAN fronthaul interface (Split 7-2). The low-level split interfaces to connect to legacy RUs are connected to Radio Signal Processer/Accelerator. This module implements the Low PHY capabilities mentioned in section 2.1.2.

The Radio signal processor can be implemented using a FPGA, DSP engine or ASICs.

CPU

CPU hosts the network operating system (NOS) which controls the transport capabilities and the Radio software which controls the radio signal processor.

Memory

DDR4 Memory is used to store the runtime data and software for the NOS and Radio software.

Storage

Non-volatile storage to store operating system, application software, firmware, operational status of Fronthaul Gateway. This could be implemented using Flash or SSD.

Timing Components

The timing module in the FHGW board is used to implement IEEE 1588 PTP and Synchronous Ethernet functionalities. This module recovers timing and provides synchronization to other modules. The module consists of below key components

1. GNSS
2. OCXO
3. DPLL
4. Servo

GNSS - Global Navigation Satellite System provides Time of the Day and Synchronization pulse (PPS) to the timing module to recover the clock and phase. This will be the primary reference clock (PRC) when selected as source of timing to the timing module

OCXO - Oven controlled crystal oscillator provides the stable reference clock to the timing module with reference to which the PTP clock is generated. This clock also is used to validate the other recovered clocks are within the required PPM offset or not

DPLL - Digital Phase lock loop circuit is used to generate the PTP clock steered by the servo algorithm. This ensures that the generated clock meets the required specifications in terms of Jitter, Phase alignment and meet the necessary masks of the ITU-T specs. This component also regenerates the required local clocks from the SyncE recovered clocks from the Phys or from the generated PTP clock or GNSS clock

Servo - This is a SW component running in the CPU or any programmable engine which analyses the timestamps from the ethernet packets, performs appropriate filtering and steers the DPLL to generate the PTP clock and Phase alignment of the clock to the primary clock source.

Ethernet PHY

This is an optional component to provide physical level ethernet connectivity. In some cases, the Network processing unit itself provides this capability and discrete ethernet PHYs are not required.

3.1.3 FHGW$_{7\text{-}2\rightarrow 8}$ Interfaces

Legacy Low-Level Split Interfaces

These interfaces are used to connect to legacy/brownfield RUs. They use protocols other than O-RAN fronthaul (Split 7-2) interface. They carry time domain Radio signals. CPRI is a typical split-8 protocol used by legacy RUs.

PCIe Interface

Support for PCIe v3 and later interface. The bandwidth depends on the use cases. It is as the control plane interface for NPUs and Radio signal processors through which the CPU controls and maintains them.

Timing Interfaces

As mentioned earlier, the timing modules provide reference timing signals to various components of the Fronthaul Gateway system. The interfaces used to provide the signals to NPU, and Radio signal processor should be specified

Ethernet Interfaces

Ethernet interfaces are used to connect to O-RUs, O-DUs and other elements in the transport network. Apart from the external interfaces, ethernet is used between the NPU and Radio signal processing unit. The bandwidth of these interfaces depends on specific use cases.

3.2 Disaggregation and Modularity Requirements

The Fronthaul gateway is implemented by combining transport and radio functions to provide both capabilities. The FHGW shall be modular enough for these two functions to be implemented separately – potentially by different vendors. This enables experts in transport and radio areas to come together and create FHGWs with flexible and programmable radio plane. In order to enable that the interfaces between transport and radio functions should be clearly defined. Below is the list of such internal interfaces.

This modularity enables multiple instances of those radio functions to be present in the FHGW. This enables increasing the scale of radio processing in the FHGW as well as integrating different Radio processing capabilities from the same different implementors.

3.2.1 Internal Interfaces

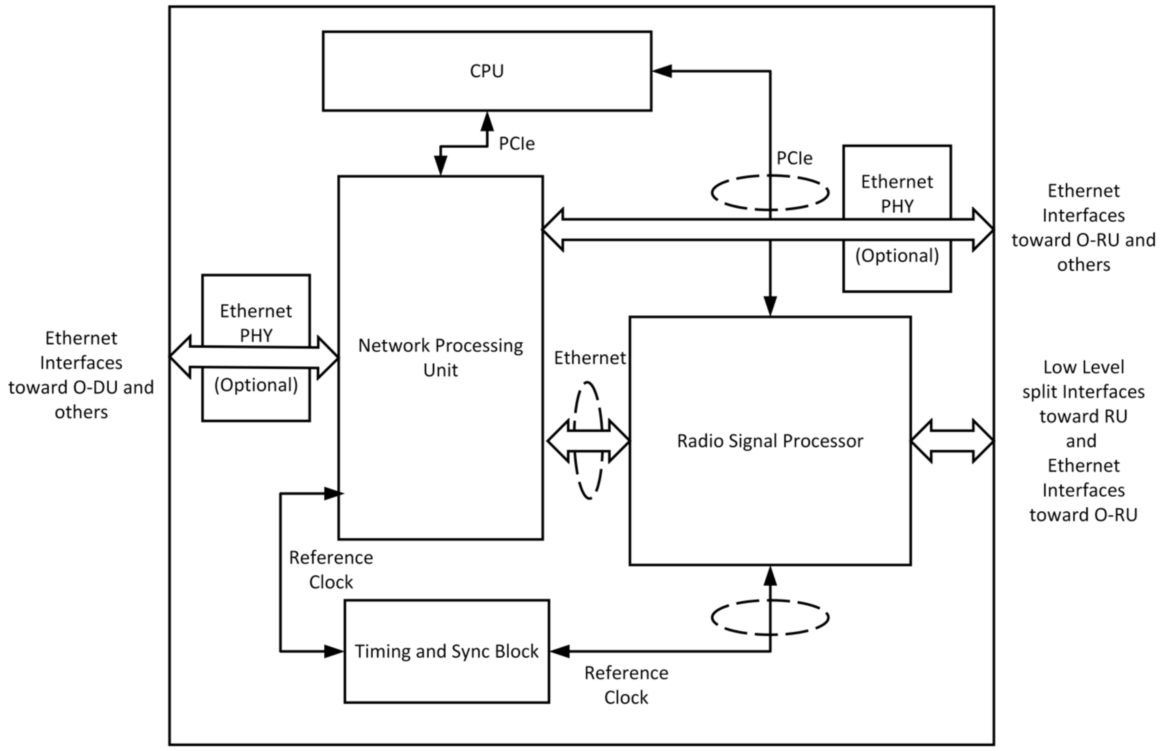

Figure 6: Fronthaul Gateway Architecture with Internal Open interfaces

Open hardware interfaces to enable a modular fronthaul gateway are given below. They are marked with dotted oval in Figure 6.

1. Datapath interface between the Radio signal processor and Network processing unit
2. Timing interface for Radio signal processor
3. Interrupt interfaces of the radio signal processor
4. Control plane access to the radio signal processor for control, maintenance and firmware management

3.3 FHGW$_{7\text{-}2\rightarrow8}$ Hardware/Performance Requirements

Table 2: FHGW$_{7\text{-}2\rightarrow8}$ Performance Requirements

Parameter	Requirement	Description	Priority
Transport capacity	FHGW shall support 300Gbps of switching	Transport capacity	High
Timing requirements	Class C compliance	Timing Requirements	High
Radio protocol translation capabilities	The FHGW shall support 240MHz of translation between O-RAN 7-2x and Split 8.	Radio BW capacity	High
Antenna Carriers	The FHGW shall support translation of 12 Antenna carriers (AxCs)	Radio Antenna capacity	High

3.3.1 FHGW$_{7\text{-}2\rightarrow8}$ Interface Requirements

Table 3: FHGW$_{7\text{-}2\rightarrow8}$ Interface Requirements

Parameter	Requirement	Description	Priority
O-RAN Fronthaul interfaces	FHGW shall support Ethernet interfaces with speeds from 10G to 25G. These interfaces will support transport of Fronthaul haul traffic compliant to O-RAN 7-2x spec: Control, User, and Synchronization Plane Technical Specification of O-RAN: O-RAN-WG4.CUS.0 [2] Below are the requirements: 6x 10G eCPRI interfaces (each port shall support 2 carriers of 20MHz i.e. total of 40MHz) 3x 25/10G eCPRI interfaces (each port shall support 1 carrier of 100MHz)	O-RAN FH Interfaces	High
Transport interfaces	FHGW shall support ethernet interfaces with speeds from 10G to 100G. These could act as O-RAN fronthaul interfaces as well. Below are the requirements: 4x25G/10G interfaces 2x10G interfaces 2x1G interfaces	Transport interfaces	High
Timing Interfaces	IEEE 1588v2 and SyncE on all ethernet interfaces. In addition, it shall/may support GNSS connectivity.	Timing Interfaces	High
Low Level Split interfaces	FHGW shall support interfaces to connect to legacy/brownfield RUs. 6x 10G legacy and non-legacy CPRI interfaces to support 1 carrier of 20MHz (capability to re-configure legacy and non-legacy CPRI interfaces to eCPRI interfaces) 3x 10G legacy and non-legacy CPRI interfaces to support 2 carriers of 20MHz i.e. total 40MHz (capability to re-configure legacy and non-legacy CPRI interfaces to eCPRI interfaces)	CPRI interfaces with rates from 2 to 8	High

Programmable Radio processor	Radio processor's parameters shall be able to be changed. FHGW shall be able to re-program radio processing function logic.	Programmability	High
Adaptability of Radio processor	The Radio processor's firmware shall be reprogrammable in the field.	Adaptability	High

Below diagram summarises the requirement given above

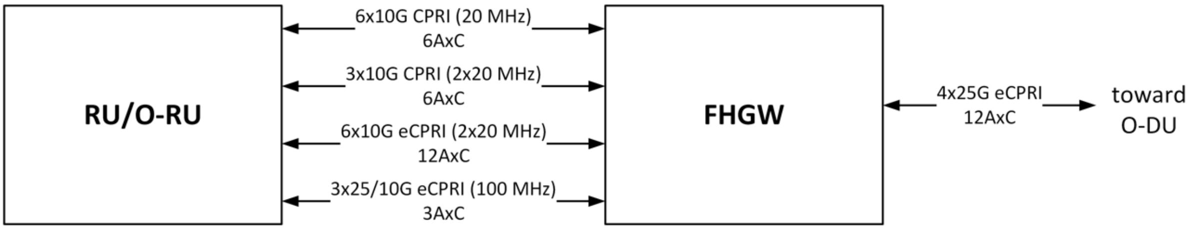

Figure 7: FHGW$_{7\text{-}2\rightarrow8}$ Interface requirements

3.3.2 FHGW$_{7\text{-}2\rightarrow8}$: Mechanical, Thermal and Power Requirements

Table 4: FHGW$_{7\text{-}2\rightarrow8}$: Mechanical, Thermal and Power Requirements

Parameter	Requirement	Description	Priority
Power Supply	-48V DC		High
Temperature	-20°C to +55°C		High

3.4 Timing Requirements

CPRI/Low Level Interface carries Radio data using a constant stream of data. Frequency and Phase are carried from the Fronthaul Gateway to the RUs using the delimiters/control characters on this bit stream. The interval of these control characters depicts the frequency, and the exact positioning of these characters carry phase information. Please refer to the CPRI specification [4] for more details.

The Fronthaul Gateway should derive these parameters from the O-DUs using IEEE 1588 PTP and SyncE or should be in sync with the O-DUs using IEEE 1588 PTP and SyncE. Then, Fronthaul Gateway should translate the above to the frequency and placement of control characters on the CPRI/Low level split interface. This way, RUs can be synchronised to the rest of the network.

3.5 Security Requirements

Fronthaul gateway merges the transport and radio functions to provide protocol translation capabilities and network capabilities. The software and firmware for transport and radio capabilities may be implemented by different vendors. So, it is important to the verify authenticity of the software and firmware used in this fronthaul gateway.

The Fronthaul gateway shall support secure storage of keys and certificates in a hardware chip and will create a secure boot and operating environment where the software and firmware loaded in it are securely signed.

Chapter 4 Hardware Reference Design

This chapter describes one example of white box hardware reference design for the fronthaul gateway when translating split option 7-2x→8 and vice versa.

4.1 FHGW$_{7\text{-}2\rightarrow8}$ Hardware Reference Design

This section describes the design of a fronthaul gateway which supports both legacy interfaces (i.e., CPRI) towards RUs and NR eCPRI interfaces toward O-RUs. When connecting to legacy RUs, it provides protocol translation to enable them to be operating with O-DUs. In addition, the FHGW provides transport capabilities for O-RAN fronthaul traffic as well as other types of traffic.

4.2 FHGW$_{7\text{-}2\rightarrow8}$ High Level Functional Block Diagram

The high-level architecture of a generic FHGW with L× Ethernet/eCPRI connections toward the O-DU, N× CPRI links toward legacy RU/RRHs, M× Ethernet/eCPRI connections toward O-RUs is shown in Figure 8. It's worth mentioning that some or all of the NxCPRI interfaces are capable of supporting O-RAN/eCPRI fronthaul interfaces through dynamic reconfiguration. Note that depending on the number of the links, the specs and number of the CPU, NPU and FPGAs used in this hardware reference design may change.

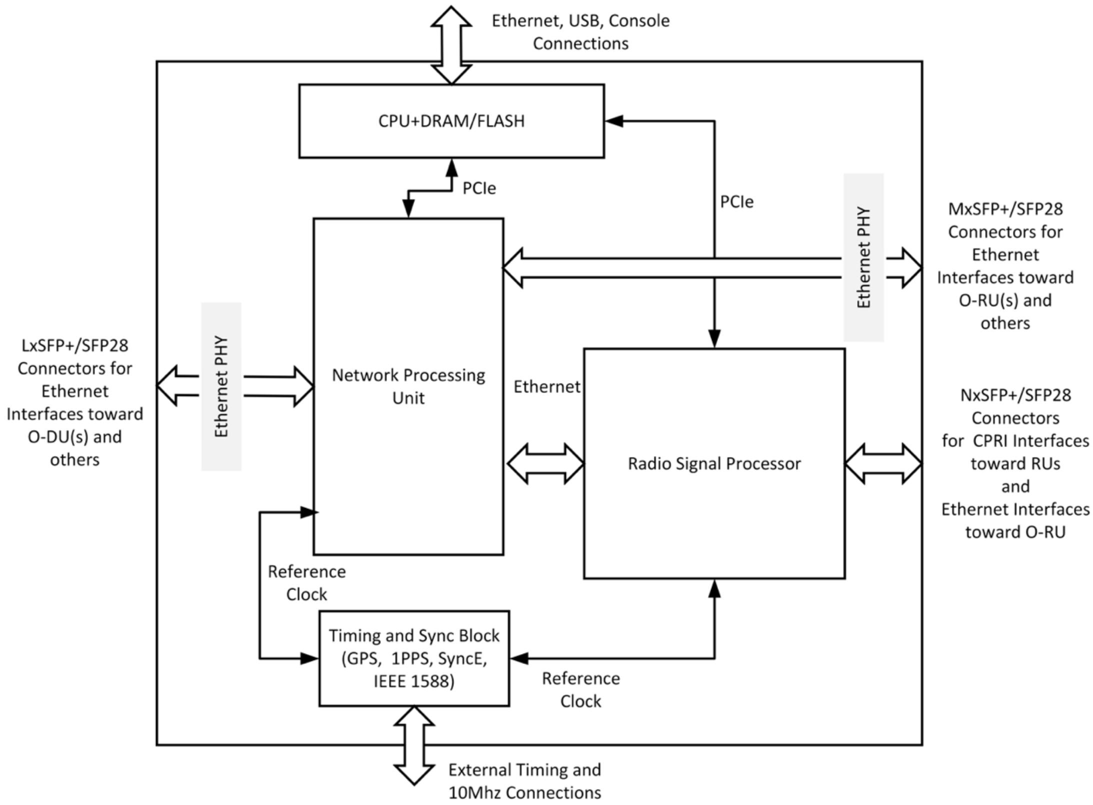

Figure 8: High-level Functional Block Diagram of the FHGW

4.3 FHGW$_{7\text{-}2\rightarrow 8}$ Hardware Components

4.3.1 CPU

The CPU runs the network operating system (NOS) for the transport functions of the FHGW and handles the control and management functions of the radio signal processor which in this example is an FPGA (including the FPGA driver). A general-purpose CPU (x86 or ARM based) with at least 8 cores can handle these functions. The requirements for the CPU are listed in the following table.

Table 5: CPU Features

Item Name	Description
Minimum number of Cores	4-8
Minimum number of Threads	8
Base Clock Frequency	1.60-1.70 GHZ
Max Turbo Frequency	1.70-2.20 GHz
Minimum L2 Cache	6-16 MB
Total Power Dissipation	17 – 25 W
Memory Types	DDR4-2666 (PC4-21300)
Maximum number of Memory Channels	2

4.3.2 Memory and Storage for CPU

Table 6: Memory and Storage Requirements

Item Name	Description
Memory	8 GB DDR4 memory
Storage	32 GB eMMC/SSD storage

4.3.3 Network Processing Unit (NPU)

The network processing unit in the FHGW handles different types of traffic, terminates Ethernet connections, and interfaces with the radio signal processing unit in order to transport data between the O-DU and legacy RUs following protocol translation in radio signal processor. The example reference design uses an NPU with the following capabilities.

Table 7: NPU Features

Item Name	Description
Switching capacity	300-410 Gbps
Network interfaces	1G, 10G, 25G, 40G, 100G
	48x10G/16x40G/4x100G
External Packet buffer (Optional)	Deep buffering, DDR4, GDDR5
OAM	Hardware based OAM
Other Features	Layer2 through Layer4 processing with integrated deep-buffer traffic management, Large on-chip forwarding tables

4.3.4 Radio Signal Processor - FPGA

Radio signal processor implements the protocol conversion between the split option 8 (CPRI) protocols used by legacy RUs and Open Fronthaul O-RAN7-2x interface (eCPRI). The functions are hosted and processed in the radio signal processor, which in this example design is an FPGA:

1) FFT/IFFT (Lower-PHY DL/UL)
2) PRACH detection (Lower-PHY UL)
3) Handling of C-Plane/M-plane messages
4) Timing and synchronization of RU
5) eCPRI framing/de-framing and switching
6) CPRI framing/de-framing and switching
7) CPRI to eCPRI conversion
8) I/Q compression on eCPRI and CPRI links

4.3.4.1 Radio signal processor – FPGA reference model 1

The physical resources (LUT, BRAM, URAM, DSP48, etc.) required on the FPGA depends on the number, speed and type of the legacy interfaces for which protocol conversion are performed. Figure 8 shows the functional block diagram of the FHGW reference design.

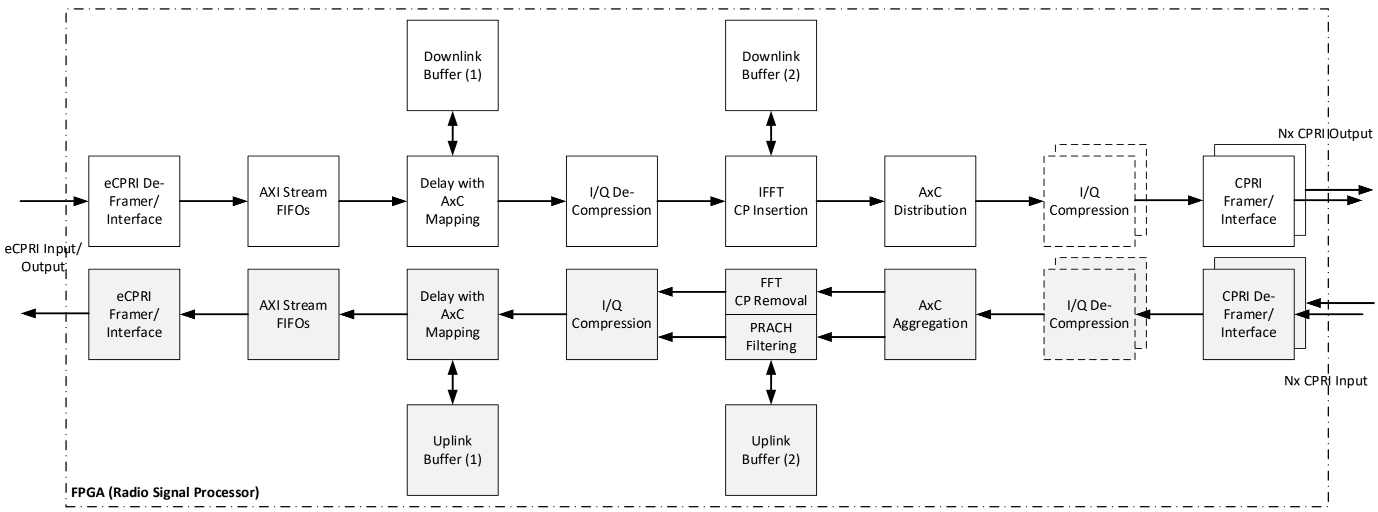

Figure 9: Radio Signal Processing Block Diagram – FPGA Model-1

4.3.4.1.1 FPGA Resource Estimation

Based on the operator requirements, we define the external connections/ports and interfaces to FHGW as shown in Figure 7.

The required number of CPRI or eCPRI [mutually exclusive] connections between the FHGW and RUs or O-RUs, respectively, would be fulfilled with 12AxC each carrier with 20MHz bandwidth for the legacy RU operation and 2x20MHz or 100MHz for the O-RU operation, which will be used in the following to calculate the FPGA resources. In this specification, we focus only on legacy RU support, thus only taking into consideration the gateway functionality of the FHGW and not the multiplexing functionality. Considering the functions that are shown in Figure 9 and the external interfaces shown in Figure 7, the reference implementation on FPGA would require hardware resources (SERDES, Flip Flops, Lookup Tables, DSPs, Block RAM, etc.) as shown in Table 8. As an example, we are assuming one LTE carrier with 20MHz of bandwidth and 1TRX antenna configuration is processed in FPGA. In time-domain, the fronthaul bandwidth of each LTE 20MHz carrier is 30.72Msps x 1 Antennas x 32bits/sample x 66/64 coding = 1.014Gbps, thus without I/Q compression, one 25Gbps fronthaul link can accommodate 24AxC. If a 2:1 ratio time-domain I/Q compression is used, 48AxC can be accommodated. Note that increasing the number of transceivers and/or the number of 20MHz LTE carriers would scale the number of required AxC.
The buffering requirements in block diagram of Figure 9 for 12AxC is 4.54Mbit and will be scaled with the number of AxC.

Table 8: Estimated FPGA Resource Allocation for the Designated Functions - 1

	Number of Instances	FF	LUT	DSP48	BRAM 18kb	URAM 288kb	Total FF	Total LUT	Total DSP48	Total BRAM 18kb	Total URAM 288kb	Notes
Sample-Rate Conversion (SRC) Filter	4	5000	3000	24	0	0	20000	12000	96	0	0	Each instance supports 3 carriers of 2T2R 20MHz at 368.72MHz clock (6 AxC); thus we need 2 instances for downlink and uplink, respectively
Block-wise Quant	2	4000	3000	12	16	0	8000	6000	24	32	0	Running at 368MHz. One instance for 6 I/Q streams (6 AxC)
Block-wise deQuant	2	3000	2000	8	10	0	6000	4000	16	20	0	Running at 368MHz. One instance for 6 I/Q streams (6 AxC)
CPRI Framing/De-framing	12	8000	3000	0	12	0	96000	36000	0	144	0	Each instance supports one 10G CPRI
Time-domain Compression/Decompression Sub-total							**130000**	**58000**	**136**	**196**	**0**	
IFFT	1	5763	3723	15	11	0	5763	3723	15	11	0	Each instance supports 6 carriers of 2T2R 20MHz at 368.72MHz clock
Add CP	1	2000	1000	0	2	6	2000	1000	0	2	6	12 OFDM symbols x 2048 samples x 32 bits x 2 dual-port memory

												read/write = 1.536Mb
FFT	1	5763	3723	15	11	0	5763	3723	15	11	0	Each instance supports 6 carriers of 2T2R 20MHz at 368.72MHz clock
CP Removal	1	2000	1000	0	2	6	2000	1000	0	2	6	12 OFDM symbols x 2048 samples x 32 bits x 2 dual-port memory read/write = 1.536Mb
Mu-law Compression	2	3000	2000	4	4	0	6000	4000	8	8	0	Mu-law compression is mainly based on look-up tables
Mu-law Decompression	2	2000	1500	4	4	0	4000	3000	8	8	0	Mu-law decompression is mainly based on look-up tables
PRACH Filtering	2	5000	4000	50	12	0	10000	8000	100	24	0	Decimate Rx signal by 16 and send to O-DU for further processing
Low-PHY Sub-total							**35526**	**24446**	**146**	**66**	**12**	

Depending on the selected CPRI line rate, the number of LTE 20MHz carriers that can be supported may vary. Table 9 shows the correspondence between the number of LTE 20MHz carriers and the CPRI line rate.

Table 9: CPRI Rate Options and the Number of Supported LTE 20MHz Carriers (AxC)

CPRI Line Rate	Line Bit Rate	Line Coding	Bits per Word	Transport Capacity (Number of 20 MHz LTE AxC Streams)
Rate-1	0.6144 Gbps	8B/10B	8	-
Rate-2	1.2288 Gbps	8B/10B	16	1
Rate-3	2.4576 Gbps	8B/10B	32	2
Rate-4	3.0720 Gbps	8B/10B	40	2
Rate-5	4.9152 Gbps	8B/10B	64	4
Rate-6	6.1440 Gbps	8B/10B	80	5
Rate-7	9.8304 Gbps	8B/10B	128	8
Rate-8	10.1376 Gbps	64B/66B	160	10
Rate-9	12.1651 Gbps	64B/66B	192	12
Rate-10	24.3302 Gbps	64B/66B	384	24

4.3.4.2 Radio signal processor – FPGA reference model 2

Figure 10 shows the top-level architecture for a reference radio signal processor. A brief description of each block is also included below.

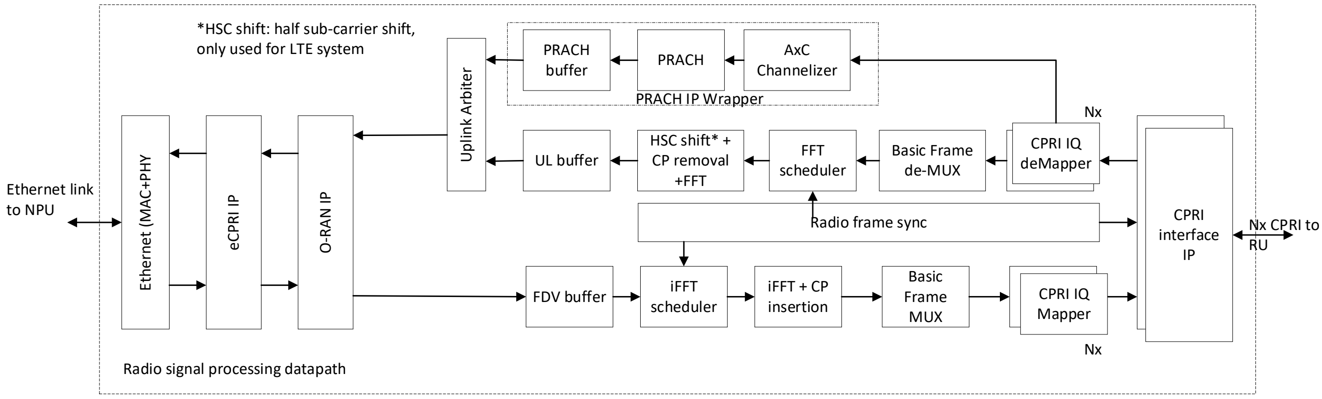

Figure 10: Radio Signal Processing Block Diagram – FPGA Model-2

4.3.4.2.1 FPGA Resource Estimation

Based on the operator requirements, we define the external connections/ports and interfaces to FHGW as shown in Figure 10. It is based on a reference implementation of a radio signal processor using different number of AxC processing capability.

The resource estimation below only considers the functionality required to provide legacy RU, or CPRI radio support. The implementation is based on the architecture depicted in Figure 10.
Considering the requirements summarized in Table 2, the number of AxCs is 12 of 20MHz LTE signals. Therefore, the O-RAN fronthaul traffic by assuming split 7.2, can be calculated as follows

ORAN traffic = [12 (AxCs) x 14 (symbols) x 1200 (REs) x 32 (bisIQ)] / 1 ms + C/S/M-plane + Pkt_header overhead ~ 6.5 Gbps

On the Ethernet link connecting to NPU or O-DU direction, one 10GE link is sufficient to accommodate the traffic. However, in our example implementation, one 25GE link is still used for scalability when more AxCs are required. Using the same calculation, one 25G link can accommodate 36 AxCs or 20MHz LTE signal using split 7.2.
Table 10 shows the estimated resource utilization (logic ALMs, DSP blocks, RAM blocks, ALUTs) for the implementation architecture from Figure 3 with respect to 12 AxCs of 20MHz signal.

Table 10: Estimated FPGA Resource Allocation for the Designated Functions - 2

	Logic (ALMs)	**DSP Blocks**	**Registers**	**RAM Blocks (M20 K)**	**ALUTs**	**comments**
25GE (soft MAC)	7,306	1	17,021	24	9,732	some FPGA devices implement the MAC in hard logic, therefore eliminating the usage for soft MAC
eCPRI IP	4,511	0	10,184	25	5,918	The eCPRI and O-RAN IP implement the eCPRI (de)framer and O-RAN (de)mapper separately. The eCPRI and O-RAN IP are dimensioned for 25GE link. More instances are needed only if more 25GE links are required. Please refer to the throughput calculation above in this subsection
O-RAN IP with (De)compression	35,352	0	78,411	15	36,527	

DL iFFT/CP	3,287	21	9,540	24	3,625	This group of blocks implement the Low PHY signal processing in DL direction. An FDV buffer is included to compensate the delay variations between FHGW and O-DU. Scale in general by the number of AxCs. The reference implementation consists of processing engines where each engine can process multiple AxCs
DL FDV buffering	6,518	0	23,550	144	2,616	
iFFT scheduler	68	0	237	0	32	
Basic Frame Mux	1,338	0	4,358	90	2,118	
CPRI IP wrapper (9 instances)	21,631	0	46,717	108	26,991	Scale by the number of CPRI interfaces
Basic frame de-mux	2,025	0	6,177	90	2,160	This group of blocks implement the low PHY signal processing in uplink including PRACH. Scale in general by the number of AxCs. The reference implementation consists of processing engines where each engine can process multiple AxCs
UL FFT/CP	3,910	27	10,723	11	4,507	
Uplink arbiter	649	0	2,102	40	452	
FFT scheduler	87	0	251	0	219	
PRACH IP wrapper	11,667	53	25,000	208	16,667	
Total	98,347	102	234,271	779	111,563	1x 25GE link, 9x CPRI links, 12 AxCs of 20MHz

4.4 Interfaces on Radio Signal Processor

4.4.1 Interface towards the NPU

The radio signal processing FPGA and the network processing unit are connected via Ethernet interfaces. The FPGA sends O-RAN 7-2x fronthaul packets to the NPU so that they can be transported to the O-DU$_{7\text{-}2}$ and the NPU sends O-RAN 7-2x fronthaul packets coming from O-DU$_{7\text{-}2}$ to the FPGA via Ethernet connection. The number and the bandwidth of these Ethernet links depends on the following conditions:

1. The number and speed (rate) of legacy CPRI interfaces toward RUs
2. The number of antenna carriers (AxC) and their respective bandwidth
3. Compression parameters on the FPGA's protocol conversion logic
4. The stream identification scheme used between FPGA and NPU
    a. All converted streams from an RU/RRH can be sent on a separate ethernet link. Simple, but uses more resources than needed.
    b. Each converted stream can be sent on a soft channel (VLAN based) on any of the Ethernet links. This requires some multiplexing and load balancing logic in both FPGA and NPU.
5. If the front panel interfaces on the FPGA needs to act as Ethernet interfaces, they need to have one to one internal interface toward the NPU. The internal interface should match the speed of front panel interface.

Depending on the above factors, there could be nx10/25G Ethernet links between the NPU and FPGA.

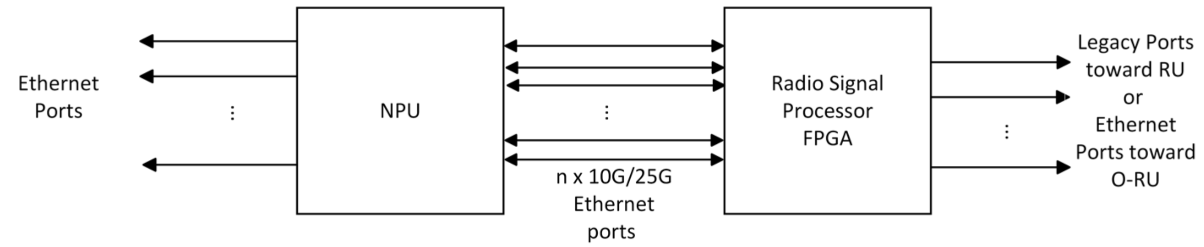

Figure 11: Internal Ethernet Links between NPU and FPGA

The management-plane traffic from the FPGA towards the radio control and management software running on the CPU also traverses through these Ethernet interfaces. The NPU will switch/route these packets towards the radio control and management software.

4.4.2 Timing and Synchronization

The timing and synchronization architecture of $FHGW_{7\text{-}2\rightarrow 8}$ looks like below.

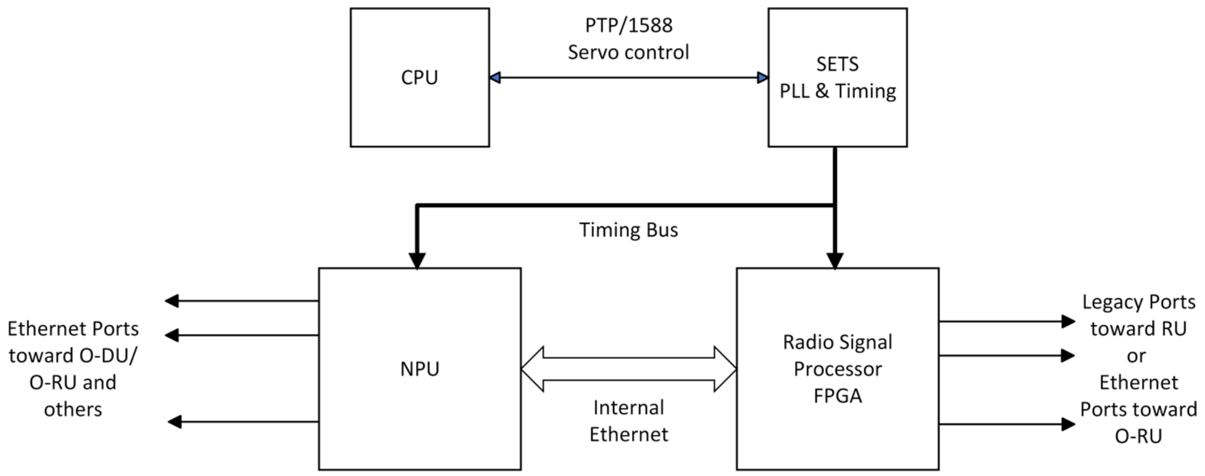

Figure 12: Timing Circuitry

The timing bus provides the following signals to the NPU and the FPGA

1. A reference clock signal to the NPU and FPGA at predetermined frequency from PTP domain.
2. A One-Pulse per Second (PPS) reference signal where the transition from low to high as sampled by the clock is the second's marker
3. A lower speed bus carrying the seconds value of Time of Day. The 48 bits of seconds value is provided serially on this signal.

4.4.3 Security

Fronthaul gateway shall include latest Trusted Platform Module (TPM) – 2.0 or greater – to store keys and certificates and to authenticate digitally signed software and firmware modules. This enables creating a secure environment where non-secure software or firmware cannot be loaded on the fronthaul gateway.

4.4.4 PCIe Interface

The FPGA is connected to the CPU via a PCIe interface for control and configuration access. During the initialization stage, the CPU detects and enumerates the FPGA as a PCIe device and sets up the access for software drivers and applications to configure, control and manage the device.

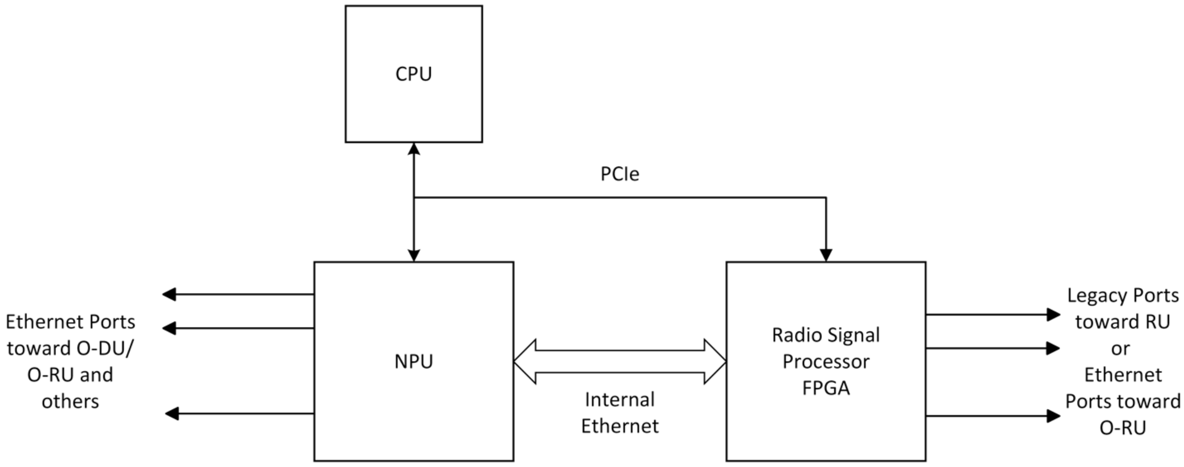

Figure 13: PCIe Interface to FPGA

4.4.5 Example Pinout of a Radio Signal Processor FPGA

Table 11: Example Pinout of the FPGA

Port	Direction	Description
DEV_TYP[1:0]	input	Device type
DEV_INST[1:0]	input	Device instance
CLK_100M_FPGA_P CLK_100M_FPGA_N	input	Free running 100MHz FPGA clock
FPGA_RST_L	input	Main Hard Reset (active low)
INTERRUPT_L	input	Main Interrupt (active low)
ETH_NPU_TX[11:0]_P ETH_NPU_TX[11:0]_N	output	Ethernet Transmit towards the NPU
ETH_NPU_RX[11:0]_P ETH_NPU_RX[11:0]_N	input	Ethernet Receive from the NPU
SFP_TX[11:0]_P SFP_TX[11:0]_N	output	CPRI 3-8 Transmit
SFP_RX[11:0]_P SFP_RX[11:0]_N	input	CPRI 3-8 Receive
ETH_REF_[3:1]_156P25M_CLK_P ETH_REF_[3:1]_156P25M_CLK_N	input	Ethernet reference clocks (156.25MHz)
CPRI_REF[4:1]_245P76M_CLK_P CPRI_REF[4:1]_245P76M_CLK_N CPRI_REF4_DUP_245P76M_CLK_P CPRI_REF4_DUP_245P76M_CLK_N	input	CPRI reference clocks (245.76MHz)
SFP[13:0]_LOS	input	SFP Loss of signal indication
PCIE_TX0_P PCIE_TX0_N	output	PCIe x1 Gen 2.0 Transmit
PCIE_RX0_P PCIE_RX0_N	input	PCIe x1 Gen 2.0 Receive

PCIE_REF_100M_CLK_P PCIE_REF_100M_CLK_N	input	PCIe FT Reference clock (100MHz)
DFPGA_PCIE_RST_L	input	PCIe reset (active high)
TOD_DFPGA_1PPS	input	One pulse per second signal
TOD_250M_CLK_P TOD_250M_CLK_N	input	1588 clock (250MHz)
TOD_IN	input	Time Of Day (seconds) value
TOD_CLKI	input	100Hz clock input used to qualify the Time-of-Day value (TOD_IN)

4.5 Scalability and flexibility of Radio Signal Processor

If multiple instances of FPGA radio signal processing functions are required to meet the requirements of a certain deployment scenario for which a single FPGA would not suffice, the modified hardware architecture shown in Figure 12 can be used. This enables scaling of radio signal processing in the FHGW as well as integrating different radio processing capabilities from the same or different implementors.
Figure 14 shows how the model can be replicated to instantiate multiple FPGAs in a fronthaul gateway to achieve higher processing capability or diverse type of processing. The FPGA instances have the same type of interfaces toward the NPU, CPU and the timing modules.

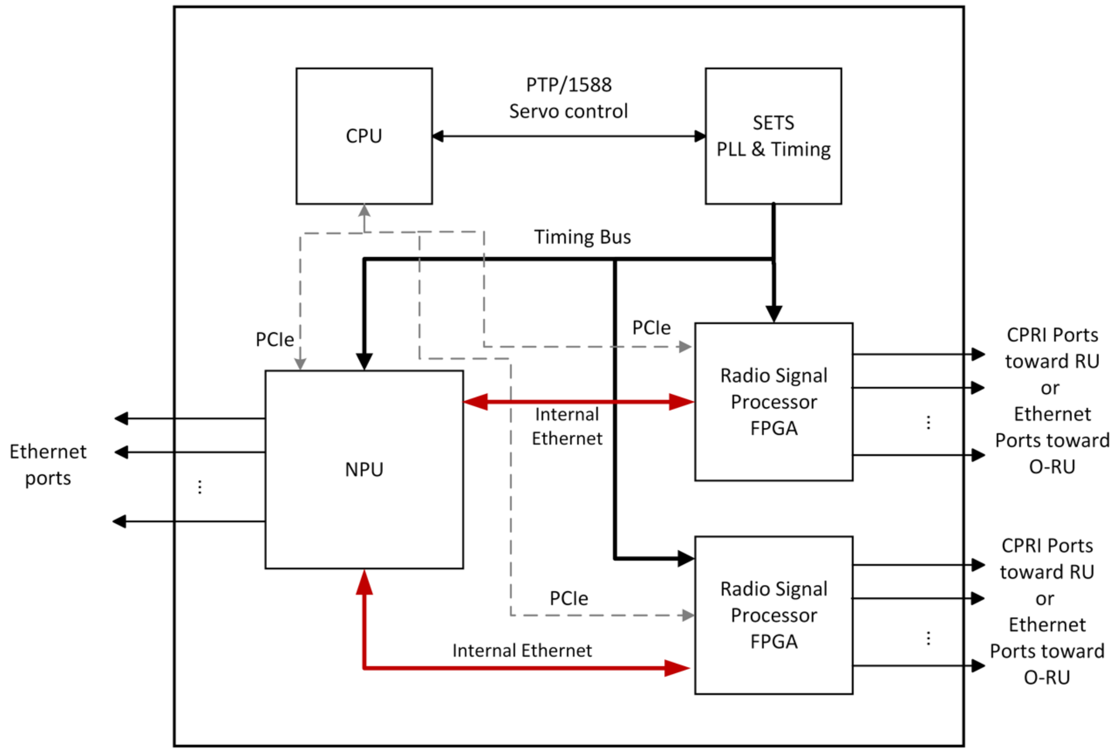

Figure 14: Multi-FPGA Architecture

4.6 External Interface Ports on the FHGW$_{7-2\rightarrow8}$

This section describes the external interface ports that are used for the operation of the FHGW$_{7-2\rightarrow8}$.

a. Hardware Requirements

The following table shows the external ports towards the RU/RRHs provided by $FHGW_{7\text{-}2\rightarrow 8}$ from the radio signal processor

Table 12: External Legacy Port List

Port Name	Feature Description
CPRI Fronthaul interface	m x 10 Gbps SFP+ transceiver interfaces

The ports on connected to NPU directly or through Ethernet PHY device(s)

Table 13: External Ethernet Port List

Port Name	Feature Description
Ethernet interfaces	n x 1 Gbps SFP+ transceiver interfaces
	p x 10 Gbps SFP+ transceiver interfaces
	q x 25 Gbps SFP+ transceiver interfaces
	r x 100 Gbps QSP28 transceiver interfaces

b. Hardware Design

- QSFP28 case and connector: The QSFP28 case and connector are standard and off-the-shelf components
- SFP+ case and connector: The SFP+ case and connector are standard and off-the-shelf components

Annex 1: Reference to Open CPRI specification

1. A reference is made to Annex 2 to Annex 7 of ORAN.WG7.IPC-HRD-Opt8.0-v02.00 [3]or succeeding revisions thereof for the example CPRI reference design in this document.
2. Note: This reference design and example implementation is solely provided as an example configuration and has no implication or bearing on the FHGW implementations and products from other vendors.

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